Day 23: File Integrity Monitoring Lab

Planned Canary-Directory Test and Expected Wazuh FIM Events

SOC Internship - TriosCyber

Field	Details
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Internship Track	SOC (Security Operations Center)
Assigned Task	Create a dedicated harmless test file in a monitored lab location, modify it, and observe the resulting Wazuh file-integrity event if the selected path is monitored. Do not alter critical system files. (Tools: Wazuh FIM, Linux/Windows endpoint)
Status of this Report	**Theoretical walkthrough. A Wazuh manager is not yet deployed in this lab, for the reasons documented in Days 17-20 and reaffirmed in Days 22 and 24. This report documents the exact syscheck configuration, the dedicated canary-directory approach, and the specific alerts that would be generated, verified against Wazuh's current official documentation and ruleset rather than reconstructed from memory. It will be re-submitted with real command output, real alerts, and screenshots once a manager is available.**

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Key Terms

- **File Integrity Monitoring (FIM):** A detection method that fingerprints (hashes) specific files or directories and alerts when that fingerprint changes, rather than inspecting file content for known-bad signatures the way antivirus does.
- **syscheck:** The name of Wazuh's FIM module; configured per-agent inside the <syscheck> block of ossec.conf, listing which directories to watch and how.
- **Scheduled Scan:** Syscheck's default monitoring mode: it walks every configured directory and compares hashes on a timer, by default every 43,200 seconds (12 hours).
- **Real-Time Monitoring:** An alternative syscheck mode (realtime="yes") that uses the operating system's own file-change notification mechanism, inotify on Linux, to detect a change within moments rather than waiting for the next scheduled scan.
- **Canary Directory:** A dedicated, disposable directory created purely to test that a monitoring pipeline works end-to-end, before extending that same monitoring to real, security-sensitive locations; the safe first step this task is built around.
- **check_all:** A syscheck directory attribute that enables the full set of integrity checks (hash algorithms, permissions, ownership, size) for that path in one setting, rather than enabling each check individually.
- **Rule 554 / 550 / 553:** Wazuh's three core syscheck alert rules: 554 fires when a new file is added to a monitored path, 550 fires when an existing monitored file's contents change, and 553 fires when a monitored file is deleted.

Important Note on This Report

This is a theoretical walkthrough, not a record of a completed FIM test. No Wazuh manager is currently reachable in this lab, for the reasons carried forward from Days 17-20, 22, and 24. Every configuration syntax, default value, and rule definition in this report was checked directly against Wazuh's current official documentation and public ruleset before being written here, rather than reconstructed from memory, following the correction made to the Day 21 and Day 24 reports.

Executive Summary

Day 23 asks for a dedicated, harmless test file to be created in a monitored location, modified, and the resulting Wazuh file-integrity event observed, explicitly avoiding any change to real system files. This depends on a running Wazuh manager and agent to actually generate and display that event, which remains unavailable in this lab for the reasons documented across Days 17-20.

This report documents the correct procedure in full: creating a dedicated canary directory rather than pointing FIM at a live system path, the exact syscheck configuration needed to monitor it in real time, the specific commands used to create and then modify a harmless test file, and the two distinct Wazuh alerts (file added, then file modified) that configuration would produce, each verified against Wazuh's genuine rule definitions rather than assumed.

1. Objective

Configure Wazuh's syscheck (FIM) module to monitor one dedicated, harmless test directory, create a test file inside it, modify that file, and record the specific Wazuh alerts generated for each action, demonstrating safe FIM testing practice by deliberately avoiding any real system or security-critical path.

2. Prerequisite: Dashboard and Agent Availability

Observing a real FIM alert requires an enrolled Wazuh agent actively reporting to a running manager, with the Discover module (covered in Day 22) available to view the resulting alert. As documented across Days 17-20 and reaffirmed in Days 22 and 24, this remains unavailable: repeated all-in-one installation attempts inside a memory-constrained VM failed at the wazuh-indexer start step, and the corrective dual-boot migration onto a shared family desktop was paused after disk preparation, since the full drive could not be backed up in the time available and proceeding with an irreversible partition change on a shared, not-fully-backed-up machine was judged too great a risk to take unilaterally. That decision stands unchanged.

The procedure below is written to be executed exactly as-is the moment a manager and enrolled agent exist.

3. Why a Dedicated Canary Directory, Not a Real System Path

The task explicitly prohibits altering critical system files, and this is worth treating as a real operational principle rather than a formality. Pointing FIM at a live, high-churn system directory (such as /etc on Linux or C:\Windows\System32 on Windows) as a first test would create two real problems: it would generate a large volume of unrelated alert noise from normal OS activity, drowning out the one test event actually being looked for, and it risks an accidental, unintended change to a file the operating system depends on. The professional practice this task mirrors is to stand up a dedicated, disposable canary directory, prove the detection pipeline works correctly end-to-end against it, and only then extend monitoring to genuinely sensitive paths with confidence that the pipeline is trustworthy.

For this lab, the canary directory is a folder created specifically and only for this test, containing no real user data or system dependencies:

```
Linux:   /home/cyberamy/fim_test
Windows: C:\FIM_Test
```

4. Planned Procedure

4.1 Create the canary directory

On Linux:

```
mkdir -p /home/cyberamy/fim_test
```

On Windows (PowerShell):

```
New-Item -Path "C:\FIM_Test" -ItemType Directory
```

4.2 Add the directory to syscheck's monitored paths

Inside the agent's ossec.conf, within the existing <syscheck> block, one <directories> entry is added for the canary path. This syntax is quoted exactly as documented in Wazuh's current official reference for the syscheck configuration block:

```
<syscheck>
  <directories check_all="yes" realtime="yes">
    /home/cyberamy/fim_test
  </directories>
</syscheck>
```

- check_all="yes": enables the full set of integrity checks for this path in one setting (hashing, permissions, ownership, and size), rather than needing to enable each check attribute individually.
- realtime="yes": switches this specific path from Wazuh's default scheduled scan, which by default only re-checks every 43,200 seconds (12 hours), to continuous monitoring via the Linux kernel's inotify mechanism (or the Windows equivalent), so a change is detected within moments rather than up to half a day later. This is the setting that makes the test practical to actually observe in one sitting.

On Windows, the same <directories> tag and attributes are used, with a Windows-style path in place of the Linux one, for example: <directories check_all="yes" realtime="yes">C:\FIM_Test</directories>.

4.3 Restart the agent to apply the change

Configuration changes to ossec.conf are not picked up automatically; the agent service must be restarted.

On Linux:

```
sudo systemctl restart wazuh-agent
```

On Windows (PowerShell, as covered in Day 20):

```
Restart-Service wazuhsvc
```

4.4 Create the harmless test file

A single, clearly-labeled, content-free test file is created inside the now-monitored canary directory.

On Linux:

```
echo "FIM lab test file - created 29 Sept 2026" >
/home/cyberamy/fim_test/test1.txt
```

On Windows (PowerShell):

```
"FIM lab test file - created 29 Sept 2026" | Out-File C:\FIM_Test\test1.txt
```

Because real-time monitoring is enabled, this file creation would be picked up by inotify almost immediately and reported to the manager, without waiting for the next scheduled scan.

4.5 Modify the test file

The same file is then modified, to generate the second, distinct event the task asks for.

On Linux:

```
echo "This line was added to trigger a FIM modification event." >>
/home/cyberamy/fim_test/test1.txt
```

On Windows (PowerShell):

```
Add-Content C:\FIM_Test\test1.txt "This line was added to trigger a FIM
modification event."
```

5. Expected Wazuh Alerts

Two separate, distinct alerts would be generated by the two actions above, each mapped here to its real Wazuh rule definition, not an assumed one.

5.1 File creation - Rule 554

Rule ID	554
Alert Level	5 (Informational / low)
Description	File added to the system.
Rule Group(s)	syscheck, syscheck_entry_added
Triggered by	Step 4.4 (creating test1.txt inside the monitored canary directory)

5.2 File modification - Rule 550

Rule ID	550
Alert Level	7 (Medium / warrants attention)
Description	Integrity checksum changed.
Rule Group(s)	syscheck, syscheck_entry_modified
MITRE ATT&CK	T1565.001 - Stored Data Manipulation
Triggered by	Step 4.5 (appending a line to the already-monitored test1.txt)

Rule 550 being a higher severity than rule 554 is itself worth noting: Wazuh treats a change to an already-existing, previously-baselined file as more significant than a brand-new file simply appearing, which reflects the real-world pattern that modification of an existing trusted file (a config, a script, a binary) is generally the more security-relevant event of the two. In this lab, both are expected and harmless, since both were deliberately self-triggered against a disposable canary file; in a real monitored environment, an analyst would still confirm who or what process made each change before closing either alert.

6. What Will Be Recorded as Evidence Once Executed

- The exact ossec.conf <syscheck> block as actually applied on the test agent, confirmed via the agent's own configuration.
- Terminal/PowerShell output of the file creation and modification commands.
- A screenshot of the Discover module (per Day 22's procedure) filtered to rule.groups: syscheck for this agent, showing both the rule 554 and rule 550 alerts with their real timestamps.
- Confirmation that no path outside the dedicated canary directory was touched during this test.

7. Conclusion

The Day 23 FIM test was not practically performed in this session, and this report does not claim otherwise. The underlying cause is unchanged from Days 17-20: no Wazuh manager or enrolled agent is currently reachable. The procedure above is built around the same safe-testing principle the task itself is checking for, using a dedicated, disposable canary directory rather than a real system path, and every configuration attribute and rule definition used here was verified against Wazuh's current official documentation and public ruleset rather than assumed.

This report will be re-submitted with the real ossec.conf change, real command output, and real rule 554 / rule 550 alerts as soon as a Wazuh manager and enrolled agent are available in this lab.